



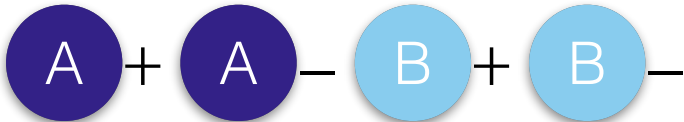
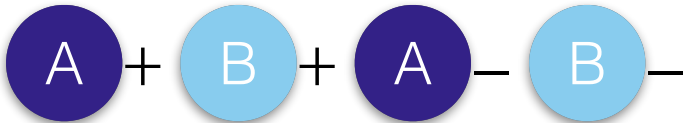
4

3



Perminituted

upd(+ -), log(+ -)

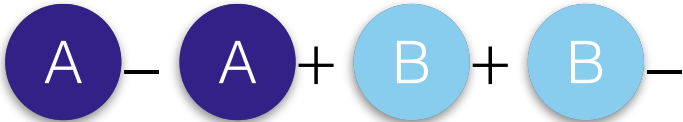


order

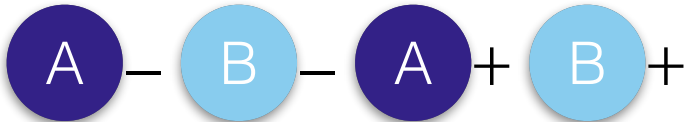
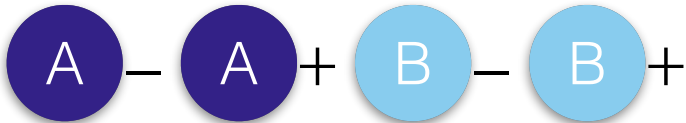
$\text{upd}(+-), \log(-+)$



$\text{upd}(-+), \text{log}(+ -)$



upd(-+), log(-+)











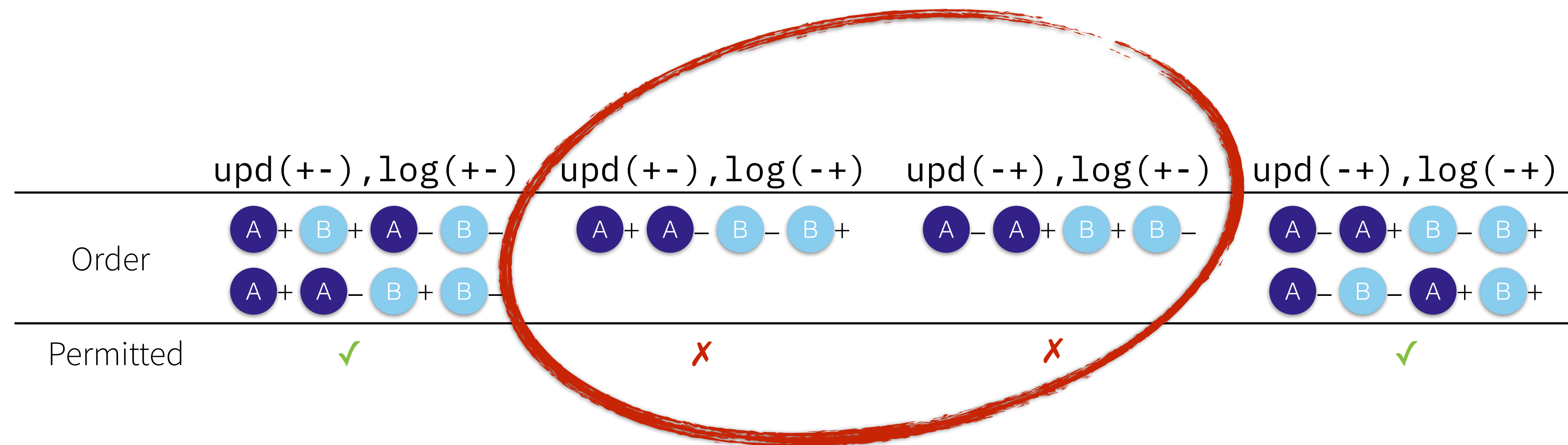


But some orders are the point!

```
def update(account: cown[Account], amount: int) {  
  when(account) {  A  
    account.balance += amount  
    val id = account.id()  
    when(log) {  B log.record(id, amount) }  
  }  
}
```

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```
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```





Ordering Design Insights

Intrinsic — Program structure determines which orders are allowed

Incidental — Scheduling and runtime must not